Internet Crimes and Cyber Security

Week: 04 **Session: 17**

Topic : Basics of Cyber Control Systems

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Basics of Cyber Control Systems

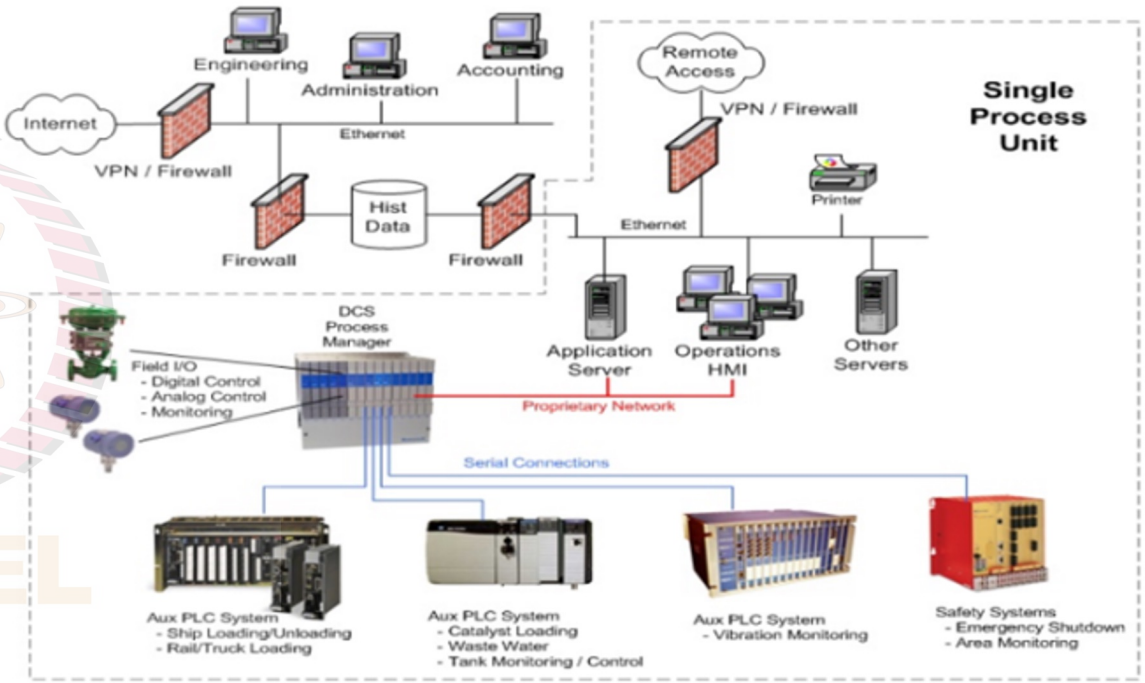
- What are Control Systems
- What are Cyber Physical Systems
- What are Industrial Control Systems (ICS)
- What are Cyber Control Systems
- SCADA Workflow
- How SCADA works as a cyber control system
- Cyber Threats to Control Systems
- Securing Cyber Control Systems
- Cyber Control Systems Standards & Regulations
- Conclusion

Basics of Cyber Control Systems

1. What are Control Systems?

- A set of devices managing, commanding, directing, or regulating behavior of systems

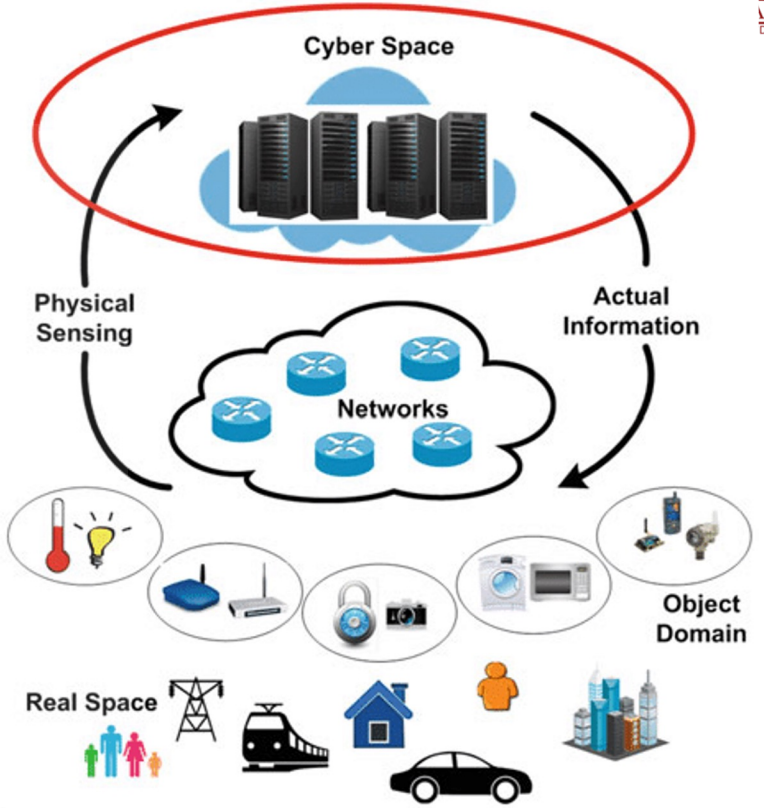
- Components:
  - Sensors
  - Controllers
  - Actuators
  - Communication Network



Basics of Cyber Control Systems

2. What are Cyber-Physical Systems (CPS)

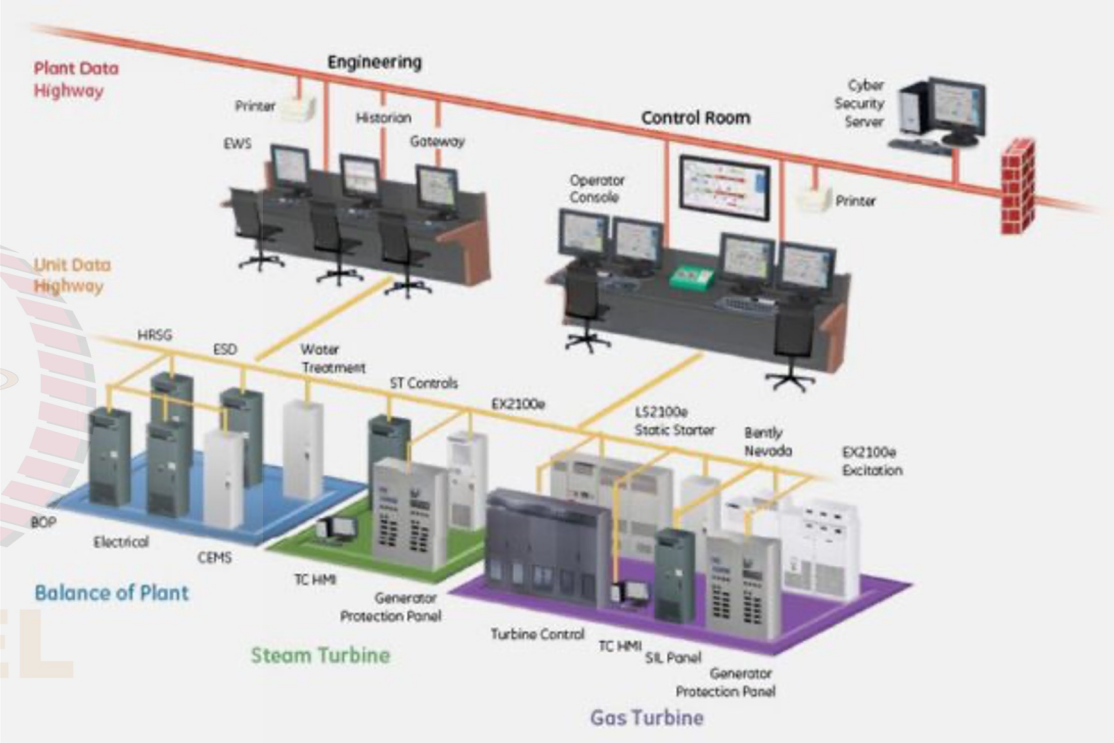
- Integration of **physical processes** with **computing and networking**
- Real-time monitoring and control
- Feedback loops between physical & digital world
- Examples:
  - Autonomous vehicles
  - Smart sensors
  - Robotics systems



Basics of Cyber Control Systems

3. What are Industrial Control Systems (ICS)

- Used in industrial automation
- Components include:
  - SCADA
  - DCS (Distributed Control Systems)
  - PLCs (Programmable Logic Controllers)
- Responsible for mission-critical operations



Basics of Cyber Control Systems

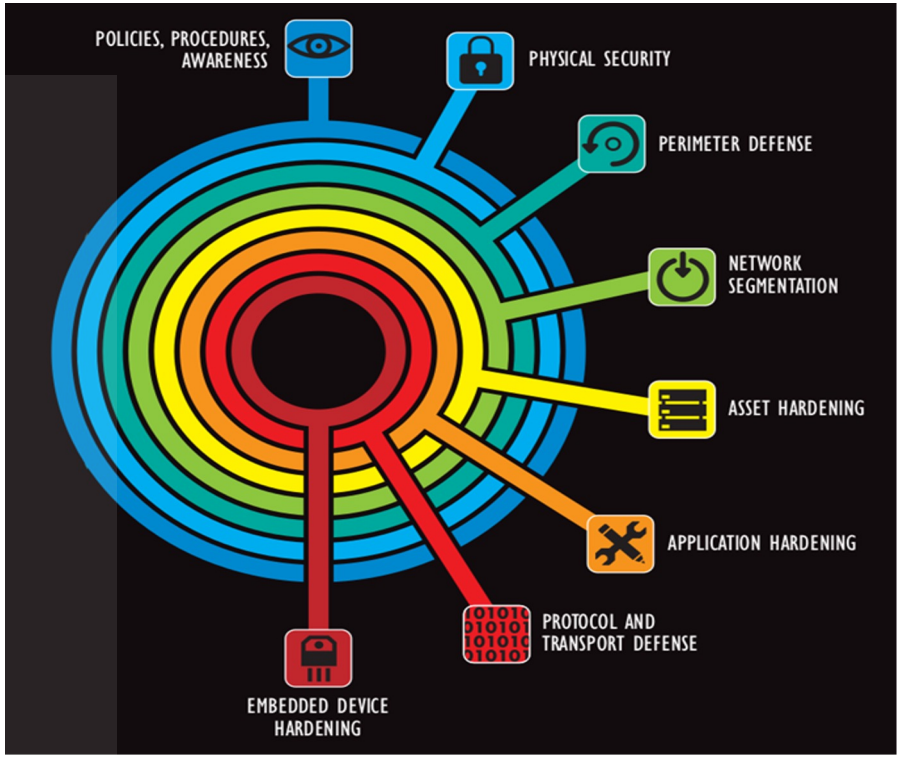
4. What Are Cyber Control Systems?

Cyber Control Systems are combinations of hardware, software, sensors, controllers, communication channels, and human-machine interfaces (HMIs) that enable automated or semi-automated control of physical processes. ‘

They include:

- **SCADA (Supervisory Control and Data Acquisition)**
- **ICS (Industrial Control Systems)**
- **DCS (Distributed Control Systems)**
- **PLC (Programmable Logic Controllers)**
- **RTUs (Remote Terminal Units)**

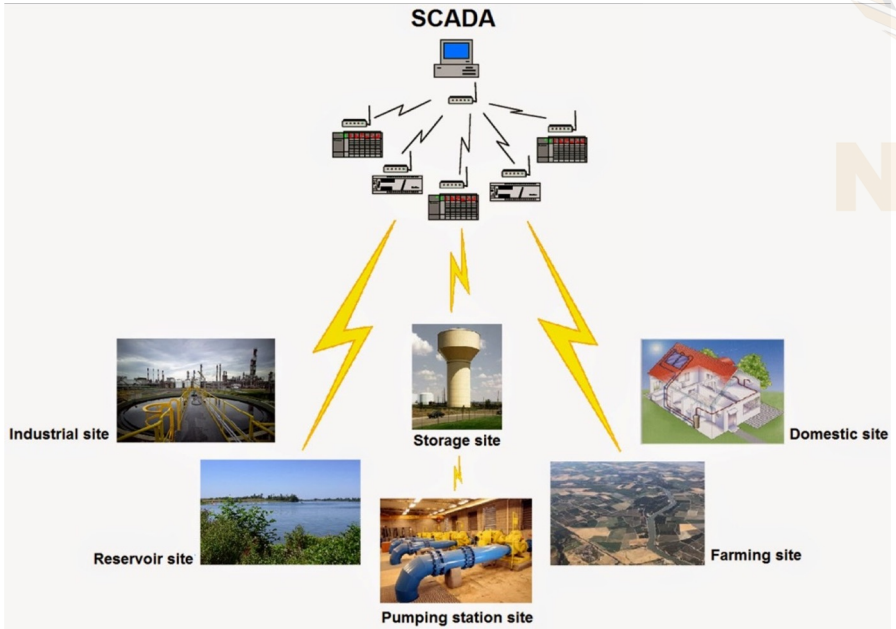
These systems collect data from the physical world, process it, and send commands to control machinery and infrastructure.



Basics of Cyber Control Systems

5. SCADA (Supervisory Control and Data Acquisition) workflow

SCADA stands for **Supervisory Control and Data Acquisition**. It is a powerful system used across industries to remotely monitor and control machines, processes, and equipment in real-time. SCADA is essential in automation industries such as manufacturing, energy, oil & gas, water treatment, and transportation.



Basics of Cyber Control Systems

6. How SCADA works as a cyber control system

- **Remote monitoring and control:** SCADA provides a central control room interface to monitor industrial processes in real-time and send commands to remote equipment like valves, motors, and pumps without on-site human intervention.

- **Data acquisition:** It collects real-time data from sensors in the field, such as temperature, pressure, or flow rates, and stores it for analysis.

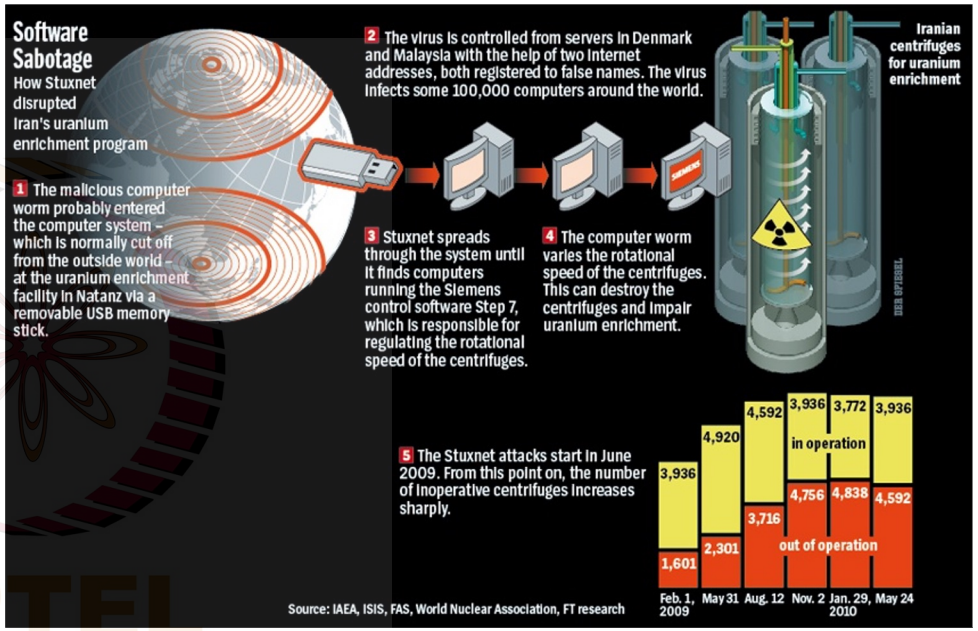
- **Bi-directional communication:** The system supports two-way communication, allowing it to both receive data from field devices and send control commands back to them.

- **Integration:** Modern SCADA systems often integrate with other technologies like the Internet of Things (IoT) and smart grids for more complex operations like load balancing and optimizing renewable energy sources.

Basics of Cyber Control Systems

7. Cyber Threats to Control Systems

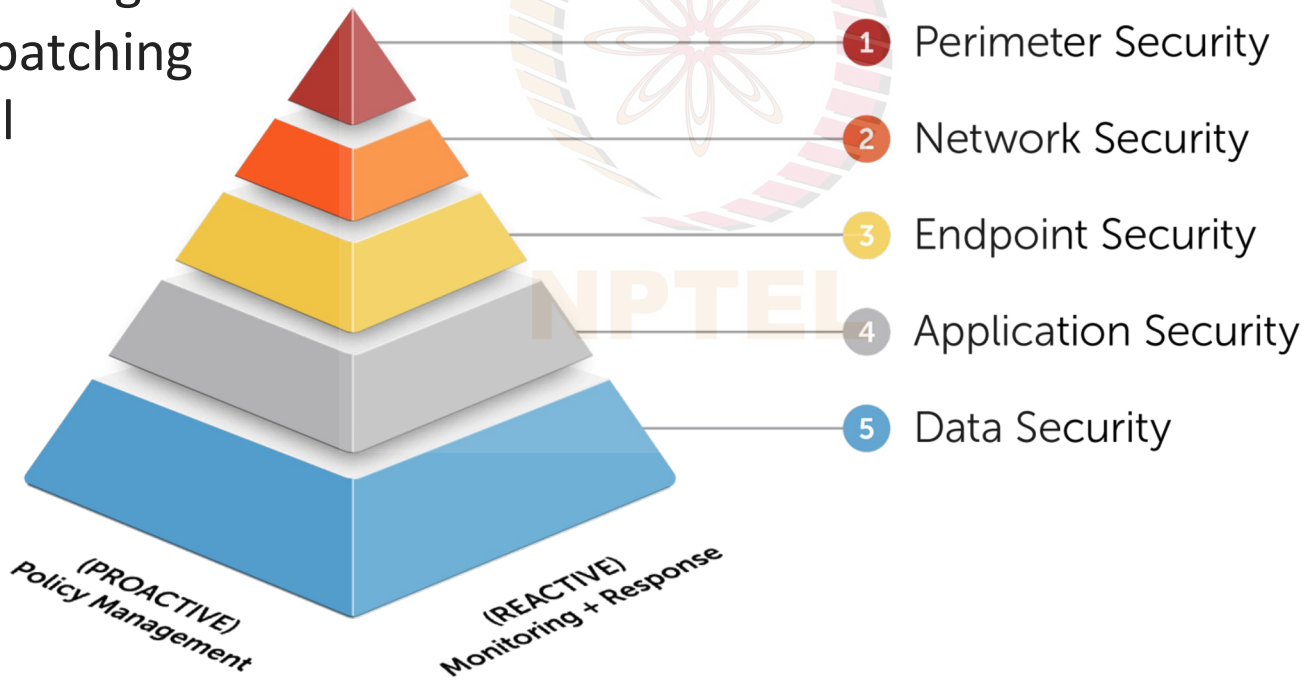
- Malware (e.g., Stuxnet)
- DoS attacks
- Man-in-the-Middle
- Unauthorized command injection
- Spoofing of sensors
- Supply chain attacks



Basics of Cyber Control Systems

8. Securing Cyber Control Systems

- Network segmentation (IT/OT separation)
- Firewalls, Intrusion Detection Systems (IDS)
- Endpoint hardening
- Vulnerability patching
- Access control



Basics of Cyber Control Systems

9. Cyber Control Systems Standards & Regulations

- NIST Framework for ICS
- IEC 62443 (Industrial Security Standard)
- ISO 27001 Information Security
- DHS ICS-CERT guidelines
- CERT advisories

Future Trends in Cyber Control Systems

- AI-driven predictive control
- Digital twins
- Autonomous industrial systems
- Edge computing
- Quantum-safe security models

Basics of Cyber Control Systems

10. Conclusion

- Cyber control systems integrate digital and physical domains
- Securing ICS/CPS is essential due to safety-critical nature
- Understanding architecture, threats, and security frameworks is key
- Future demands: intelligent, resilient, adaptive cyber-controls